British Journal of Clinical Psychology (2017)

The British Psychological Society

www.wileyonlinelibrary.com

Don't worry, be happy: The role of positive emotionality and adaptive emotion regulation strategies for youth depressive symptoms

Marie-Lotte Van Beveren[1]*, Kaitlin Harding[2], Wim Beyers[1] and Caroline Braet[1]

[1]Department of Developmental, Personality and Social Psychology, Ghent University, Belgium

[2]Department of Clinical Psychology, Seattle Pacific University, USA

Objectives. Low positive emotionality (PE) represents a temperamental vulnerability to depression in youth. Until now, little research has examined the mechanisms linking PE to depressive symptoms. Starting from integrated cognitive-affective models of depression, we aimed to study adaptive emotion regulation (ER) as a key underlying mechanism in the temperament–depression relationship.

Methods. This study investigated whether adaptive ER strategies mediate the association between PE and depressive symptoms in a large community-based sample of youth, using a cross-sectional design. Participants were 1,655 youth (54% girls; 7–16 years, M = 11.41, SD = 1.88) who filled out a set of questionnaires assessing temperament, adaptive ER strategies, and depressive symptoms.

Results. Results revealed that low PE was significantly related to higher depressive symptoms among youth and that a lack of total adaptive ER abilities mediated this relationship. More specifically, the infrequent use of problem-solving appeared to be of significant importance. Problems in positive refocusing and a deficient use of forgetting mediated the relationships between low PE and high negative emotionality (NE) in predicting depressive symptoms. Reappraisal and distraction were not significant mediators.

Conclusion. Results highlight the need to account for temperamental PE and adaptive ER strategies when studying youth depression. The findings contribute to a more nuanced understanding on the differential role of temperamental risk factors for developing depressive symptoms at an early stage and advocate for greater attention to adaptive ER strategies.

Practitioner points

- Clinical interventions for youth depression may be improved by incorporating adaptive emotion regulation (ER) strategies and enhancing positive emotions.
- Youth low in positive emotionality (PE) may especially benefit from learning adaptive ER skills.

*Correspondence should be addressed to Marie-Lotte Van Beveren, Department of Developmental, Personality and Social Psychology, Ghent University, Henri Dunantlaan 2, Ghent 9000, Belgium (email: marielotte.vanbeveren@ugent.be).

DOI:10.1111/bjc.12151

- Clinical practitioners should focus on alleviating negative emotions and enhancing positive emotions, especially among youth low in PE.

Depression is a debilitating disorder characterized by sad mood and the inability to experience positive affect or pleasure (i.e., anhedonia; American Psychiatric Association, 2013). Prevalence rates of depression markedly increase from 1% in childhood to 8% in adolescence, with a lifetime prevalence in adolescents of 15% to 20% (Birmaher *et al.*, 1996). In Europe, approximately 10.5% youth meet criteria for clinical depression and over 29% report subclinical depressive symptoms (Balazs, 2013). Youth depression is associated with a number of undesirable outcomes (Burcusa & Iacono, 2007; Copeland, Shanahan, Costello, & Angold, 2009; Fergusson & Woodward, 2002), including a sixfold increased risk for attempting suicide (Nock *et al.*, 2013). These findings underline the necessity to prioritize research on predictors of depressive symptoms during this critical developmental period.

Inspired by affective models of depression, research has recently started to address the role of reactive temperament traits in predicting youth depression. In particular, high *negative* emotionality (NE) and low *positive* emotionality (PE) are crucial risk factors for the development of youth depression (Lonigan, Phillips, & Hooe, 2003). Integrated cognitive-affective models of depression have implicated several emotion regulation (ER) strategies as mediating mechanisms in the developmental translation from these temperament traits to depression. Until now, research has largely focused on the role of *maladaptive* ER strategies as risk factors for youth depression (Mezulis, Priess, & Hyde, 2010) and the greater presence of maladaptive ER strategies among youth with high NE (Mezulis *et al.*, 2010). However, recent studies suggest that a lack of *adaptive* ER strategies also may confer risk of depression (Braet *et al.*, 2014; Schäfer, Naumann, Holmes, Tuschen-Caffier, & Samson, 2016) and impair youth's ability to adaptively regulate negative affect, which is uniquely related to temperamental PE (Van Beveren, McIntosh, *et al.*, 2016; Van Beveren, Mezulis, Wante, & Braet, 2016). This study seeks to integrate the largely distinct empirical discussions on temperament and ER by examining the joint contributions of PE and adaptive ER strategies on youth depressive symptoms.

PE and depression

Temperamental models of depression assume that life events elicit negative and positive emotions distinctly as a result of reactive temperament (Derryberry & Rothbart, 1997). Reactive temperament can be defined as the excitability, responsivity, or arousability of the behavioural and physiological systems of the organism (Derryberry & Rothbart, 1997). Temperamental theories typically identify NE and PE as two distinct, developmentally stable and reactive temperament traits. NE has been studied extensively in the development of youth depression and refers to the susceptibility to be affected by negative emotions. NE involves features such as sadness, anger, anxiety, irritability, and negative mood (Rothbart, Ahadi, & Evans, 2000) and is assumed to encompass the constructs of negative affectivity (Clark & Watson, 1991), the behavioural inhibition system (BIS; Corr, 2002), and neuroticism (Eysenck, 1967). Depression research also recognizes the role of PE as a reactive temperament trait. PE refers to the susceptibility to be affected by positive emotions and involves features such as reward-seeking behaviour, sociability, and positive mood (Derryberry & Rothbart, 1997). In contrast to NE, PE comprises the constructs of positive affectivity (Clark & Watson, 1991), the behavioural activation system (Corr, 2002), and extraversion (Eysenck, 1967).

Starting from affective models on depression, high NE and low PE are identified as vulnerability factors for the development and maintenance of depressive symptoms in children, adolescents, and adults (Clark & Watson, 1991; Lonigan *et al.*, 2003). Although both theory and research suggest that low PE is also of specific interest when studying depression (Lonigan *et al.*, 2003; Van Beveren, McIntosh, *et al.*, 2016; Van Beveren, Mezulis, Wante, & Braet, 2016), research on affective vulnerabilities to depression focuses on NE and largely ignores the role of PE for depression (Gilbert, 2012). Yet, as the effects of positive emotions appear to be distinct from those of negative emotions (Garland *et al.*, 2010), a more profound insight into problems in positive emotional functioning may provide a promising avenue for better understanding the risk of youth depression (Carl, Soskin, Kerns, & Barlow, 2013). Unfortunately, to date less is known about the mechanisms by which PE exerts its effect on youth depressive symptoms.

Integrated cognitive-affective models on depression

Cognitive behavioural theories on depression assert that an individual's cognitive and behavioural responses to life events may diminish or enhance the emotions elicited by these events (Beck, 1970; Lewinsohn, 1975). In addition, the broaden-and-built theory specifies that, unlike negative emotions which narrow people's thought–action repertoires, positive emotions broaden one's cognitive abilities and behavioural repertoires (Fredrickson, 2001). Over time, these cognitive and behavioural responses constitute stable ER strategies that influence the experience of negative and positive emotions and buffer against negative emotional experiences (Tugade & Fredrickson, 2007). The broaden-and-build theory presumes a link between positive emotions and adaptive ER strategies. Literature on the beneficial effects of positive emotions for emotional functioning in youth is less extensive compared to research on adult depression. However, emerging evidence supports the application of the broaden-and-build theory among younger age groups, providing an interesting framework from which to understand the role of positive emotions in youth (Gilbert, 2012). In following an integrated cognitive-affective model, this study proposes a lack of adaptive ER strategies as one possible pathway linking low PE to youth depressive symptoms.

Poor ER of negative emotions is common among depressed individuals (Henry, Castellini, Moses, & Scott, 2016; Mehrabi, Mohammadkhani, Dolatshahi, Pourshahbaz, & Mohammadi, 2014). ER refers to the set of processes by which individuals influence which emotions they have, when they have them, and how they experience and express these emotions (Gross, 1999). ER can be effortful or non-effortful, conscious or unconscious (Gross & Thompson, 2007), and can take place intrapersonally or interpersonally (Marroquín, 2011). While interpersonal ER influences risk of depression (Marroquín, 2011), this study focuses on intrapersonal ER strategies given their stronger relationship to depression (Carl *et al.*, 2013). From a theoretical perspective, certain ER strategies are considered *maladaptive* because they are associated with greater overall psychopathology in the long term (Aldao, Nolen-Hoeksema, & Schweizer, 2010; Garnefski & Kraaij, 2006). For example, rumination is maladaptive in regulating negative emotions (Hyde, Mezulis, & Abramson, 2008) as this strategy has repeatedly predicted the onset, severity, persistence, and recurrence of depressive symptoms in children, adolescents, and adults alike (for a review see e.g., Nolen-Hoeksema, Wisco, & Lyubomirsky, 2008). On the other hand, strategies are labelled *adaptive* if they predict greater emotional well-being (Aldao *et al.*, 2010). For instance, problem-solving as an adaptive strategy promotes emotional functioning by eliminating or altering stressors that elicit negative emotions (Aldao *et al.*, 2010; Folkman & Moskowitz, 2000). The growing field of research urges to pay further attention to deficits in adaptive ER for explaining the risk of developing psychopathology (Aldao *et al.*, 2010; Schäfer *et al.*, 2016). Unfortunately, few studies have researched the role of adaptive ER strategies for youth depression.

Adaptive ER strategies

Several ER strategies are supported as adaptive in regulating negative emotions in youth and adult samples. Adaptive ER strategies related to depression include reappraisal, problem-solving, acceptance, positive refocusing, distraction, and forgetting (Aldao *et al.*, 2010; Braet *et al.*, 2014; Garnefski & Kraaij, 2006; Garnefski, Kraaij, & Spinhoven, 2001).

Reappraisal and problem-solving are two ER strategies that are adaptive across situations (Aldao *et al.*, 2010; Nolen-Hoeksema, 1991). Reappraisal, reinterpreting the meaning or value of a negative event, is an essential component of cognitive behavioural therapy (CBT; Beck, Rush, Shaw, & Emery, 1979) and has been shown to reduce negative emotions (Samson & Gross, 2012). In addition, this strategy is supported to increase positive affect and facilitate adaptive ER (Tugade & Fredrickson, 2007). Although reappraisal has repeatedly proven its effectiveness, it requires considerable cognitive resources for youth to adopt during emotionally charged situations (Hannesdottir & Ollendick, 2007). Furthermore, CBT programmes only appear to work for some youth (Hannesdottir & Ollendick, 2007). As a result, researchers and clinicians value the incorporation of alternative adaptive ER strategies into current treatment programmes (Hannesdottir & Ollendick, 2007). Some CBT treatments also include problem-solving training (Stark & Kendall, 1996). Problem-solving refers to the cognitive and behavioural processes by which an individual attempts to identify or discover effective or adaptive solutions for stressful problems (D'Zurilla & Nezu, 1999). Problem-solving has been shown to protect youth from developing depressive symptoms (Hilt, McLaughlin, & Nolen-Hoeksema, 2010) and to increase positive affect (Folkman & Moskowitz, 2000).

Acceptance is a third adaptive strategy that empirical research has taken growing interest in. The purpose of acceptance is to become aware of feelings, observe them as they are, and to accept them without judging them (Williams, Teasdale, Segal, & Soulsby, 2000). Research suggests that acceptance promotes improved mental health outcomes (Heffner, Eifert, Parker, Hernandez, & Sperry, 2003). Furthermore, acceptance is an important component in mindfulness-based therapies (Teasdale *et al.*, 2000), which have been demonstrated to improve psychological functioning in adolescents (Bogels, Hoogstad, van Dun, de Schutter, & Restifo, 2008; Van der Gucht, Kuppens, Maex, & Raes, 2016) and to promote positive emotions (Jimenez, Niles, & Park, 2010).

Positive refocusing, distraction, and forgetting also are proposed as protective strategies against youth depression, yet these ER strategies have received less attention. Positive refocusing refers to thinking about joyful and pleasant issues instead of thinking about the actual stressful event (Garnefski & Kraaij, 2006). Positive refocusing is negatively associated with depressive symptoms in adolescents (Braet *et al.*, 2014; Garnefski & Kraaij, 2006) and has been demonstrated to increase positive affect (Folkman & Moskowitz, 2000). While avoidance ER strategies (e.g., withdrawal) have been repeatedly identified as risk factors for overall psychopathology, distraction and forgetting are not classified as maladaptive ER strategies. Both ER strategies are seen as adaptive because they involve active problem-solving strategies (Grob & Smolenski, 2005). Distraction constitutes the intentional deployment of attention away from one's negative emotions or self-related stress to an external situation or stimulus (Gross, 1999) and has been found to be an adaptive ER strategy across various studies (Sheppes, Scheibe, Suri, & Gross, 2011). However, the hypothesized beneficial effects of distraction on depressive symptoms have not been clearly demonstrated (e.g., Just & Alloy, 1997; Nolen-Hoeksema & Morrow, 1993). Furthermore, distraction may have limited advantages short-term and adverse consequences long term (Gross, 1998; Van Dillen & Papies, 2015). Lastly, forgetting is a cognitive strategy referring to an individual's effort to intentionally forget about the event that is eliciting negative emotions. Forgetting is supported to be a powerful adaptive ER strategy, as forgetting the bad (and remembering the good) is strongly associated with greater well-being and overall life satisfaction (Charles, Mather, & Carstensen, 2003; Kennedy, Mather, & Carstensen, 2004).

While most research has studied the role of ER strategies for depression in isolation, it is unlikely that youth make use of only a single ER strategy for regulating their everyday feelings. Rather, multiple ER strategies often are employed in different contexts and to different degrees within one individual (Aldao & Nolen-Hoeksema, 2012; Brans, Koval, Verduyn, Lim, & Kuppens, 2013) and different ER strategies may be used concurrently (Gross & Thompson, 2007). Therefore, it is important to consider the total range of ER strategies when examining the role of ER on psychological functioning (Brans *et al.*, 2013; Yap, Allen, & Sheeber, 2007).

Adaptive ER strategies may mediate the effect of PE on depression

Emotion regulation is supported as an underlying mechanism in the temperament–depression relationship (Hyde *et al.*, 2008; Yap *et al.*, 2007), but only a handful of studies have explored the relationship between low PE, adaptive ER strategies, and youth depressive symptoms. Preliminary evidence in young adults suggests that deficits in adaptive ER abilities may represent an underlying mechanism in the relationship between low PE and greater depressive symptoms (Harding, Hudson, & Mezulis, 2014; Hudson, Harding, & Mezulis, 2015). However, to our knowledge only two studies have explored the joint contributions of reactive temperament traits and adaptive ER for psychopathology in younger age groups. One study revealed that an adaptive ER style significantly mediated the relationship between PE and hypomanic symptoms in primary school children (Verstraeten, Vasey, Raes, & Bijttebier, 2012). Moreover, Yap *et al.* (2011) found that self-reported adaptive ER responses to sadness partially mediated the association between NE and depressive symptoms in early adolescents. However, research has hitherto neither examined the mediating role of the total range of adaptive ER abilities nor has research clarified the mediating role of the different adaptive ER strategies in the PE–depressive symptoms relationship in youth. To address this gap, we sought to examine the collective and distinct effects of adaptive ER abilities while accounting for the reactive temperament trait of NE.

Current study

This study investigated the unique associations between temperament, adaptive ER strategies, and depressive symptoms in a large community-based sample of youth, using a cross-sectional design. We hypothesized that (1) lower levels of PE (and higher levels of NE) would be associated with higher levels of depressive symptoms, (2) that lower total adaptive ER abilities would be associated with higher levels of depressive symptoms, and (3) that a lack of specific adaptive ER strategies would be associated with higher symptom levels. In following integrated cognitive-affective models of depression, we hypothesized that (4) total adaptive ER abilities would mediate the relationship between PE and depressive symptoms, but not the relationship between NE and such symptoms. Finally, (5) we hypothesized that all adaptive ER strategies would contribute to understanding the PE–depressive symptoms relationship. Given the gender and age differences in adolescents' ER and depressive symptomatology (Hyde *et al.*, 2008; Thayer, Rossy, Ruiz-Padial, & Johnsen, 2003), we included these variables as covariates in analyses. We also aimed to explore the moderating role of these variables through mediation. Based on theoretical and empirical literature, we hypothesized that the proposed relationships between reactive temperament traits and adaptive ER (6) would become stronger with increasing age (McRae *et al.*, 2012; Steinberg, 2005) and (7) would be stronger for girls than boys (Hyde *et al.*, 2008). We controlled for the effects of the regulative temperament trait (effortful control (EC)) in analyses as research has shown that all temperament traits should be considered jointly to fully understand the distinct role of one specific temperament trait for depression (Van Beveren, Mezulis, Wante, & Braet, 2016; Vasey *et al.*, 2014).

Method

Participants

Participants were 1,646 Dutch-speaking adolescents (54% girls) ranging from 7 to 16 years (M = 11.41, SD = 1.88). Participants were recruited from all public schools in a normal urban city in Belgium (Deinze).

Procedure

This study is part of a larger research project on depression, including all children from 4th to 8th grade from all public schools of an urban city (Deinze, Belgium). Further details can be found in Van Beveren, Mezulis, Wante, & Braet (2016) For this study, data from the third wave were used. Briefly, youth were sent home with an informational letter describing the aim of the study, inviting them to take part, and asking for parental permission prior to participation. After written consent, all youth completed questionnaires in their classroom in the presence of a teacher during regular school hours. The questionnaires were administered in a fixed order. Prior to the start of the assessment, teachers were informed about the aim of the study and provided clear instructions to guide the classroom assessment to ensure standardization. Participation was not remunerated.

Measures

Reactive temperament

The trait version of the Positive and Negative Affectivity Schedule for Children (PANAS-C; Laurent *et al.*, 1999) was used to assess *reactive* temperament. The PANAS-C is a self-report instrument that contains emotion items. Participants rated the extent to which they *usually* experience each specific emotion on a 5-point Likert scale. The questionnaire consists of two subscales, each containing 15 items, that assess NE and PE. The PANAS-C demonstrated good convergent and discriminant validity with child self-reports of depression and anxiety (Laurent *et al.*, 1999). Cronbach's alphas for the NE and PE subscale were .82 and .81, respectively.

Regulative temperament

Effortful control was assessed using the Early Adolescent Temperament Questionnaire – Revised (EATQ-R; Ellis & Rothbart, 2001; Hartman, 2000). The EATQ-R is designed to measure temperament traits in children age 9–15. In this study, a shortened version of the EATQ-R was used, assessing only the EC factor. Responses were rated on a 5-point Likert scale from (1) *almost always untrue* to (5) *almost always true*. The EC factor is a combination of three subscales: Attention (six items), Inhibitory control (five items) and Activation Control (five items) scales. The EATQ-R EC scale demonstrated acceptable internal consistency and good test–retest reliability among non-clinical youth (Muris & Meesters, 2009). Cronbach's alpha in this study was .78.

Adaptive emotion regulation

Adaptive ER strategies were measured using the FEEL-KJ (Braet, Cracco, Theuwis, Grob, & Smolenski, 2013; Grob & Smolenski, 2005). The FEEL-KJ is a 90-item self-report measure assessing various adaptive, maladaptive, and external ER strategies in response to anger, fear, and sadness in youth aged 8–18 years. Participants rated each item on a 5-point Likert scale from (1) *almost never* to (5) *almost always*. In this study, participants only filled in the items for the *adaptive* strategies (AS) scale which includes seven strategies: *reappraisal, behavioural problem-solving*, *cognitive problem-solving, acceptance, positive refocusing, distraction*, and *forgetting*. Total scores on the adaptive ER scale (FEEL-KJ-AS) comprise all three emotions and reflect general dispositions to adaptively cope with negative emotions. The FEEL-KJ has been proven to be a valid and reliable questionnaire (Braet *et al.*, 2013; Schmitt, Gold, & Rauch, 2012). Internal consistency for the FEEL-KJ-AS in this study was .94. Internal consistency for the separate subscales ranged from acceptable to good with reappraisal (.87), problem-solving (.76), acceptance (.72), positive refocusing (.79), distraction (.87), and forgetting (.76). The behavioural problem-solving and the cognitive problem-solving subscales were merged into one overarching problem-solving scale to adopt a more parsimonious approach.

Depressive symptoms

The Child Depression Inventory (CDI; Kovacs, 1992; Timbremont & Braet, 2002) was used to measure depressive symptoms. The CDI is a 27-item self-report measure for youth aged 7–17 years to assess cognitive, affective, and behavioural symptoms of depression. Each item comprises three response options, which vary in severity and are rated on a 3-point scale. The CDI demonstrated good internal consistency and test–retest reliability in non-clinical youth (Craighead, Smucker, Craighead, & Ilardi, 1998; Timbremont & Braet, 2002). Cronbach's alpha in the current sample was .85.

Data analytic strategy

Missing values

Preliminary analyses of the data suggested that the percentage of missing data ranged between 0% and 3.8% per item. Comparison of means and covariances of all variables using Little's (1988) MCAR test produced a normed χ^2 (χ^2/df) of 1.23, $p > .20$, indicating that the data were likely missing completely at random (Bollen, 2014). Consequently, missing values were estimated following the expectation–maximization (EM) algorithm available in SPSS 23 (Schafer, 1997).

Data analytic plan

To adjust for measurement error, structural equation modelling with latent variables (*SEM*; Bollen, 2014) was used to test the proposed relationships using robust maximum likelihood estimation in Mplus 7.6 (Muthén & Muthén, 2012). *SEM* with latent variables requires multiple indicators for all study variables. Variables were modelled as latent variables, each represented by three parcels and created through the random selection of items. Total adaptive ER was modelled as a higher order factor indicated by the specific adaptive ER strategies. Given the lack of an established factor structure in the EATQ-R (Ellis & Rothbart, 2001; Muris & Meesters, 2009; Putnam, Ellis, & Rothbart, 2001), the EC subscale of the EATQ-R was not modelled as a latent variable. Instead, we used the total scale score as a manifest variable in all analyses.

Model fit was evaluated based on the combined cut-off of .06 for the root mean square error of approximation (RMSEA) and .08 for the standardized root mean square residual (SRMR). In addition, a comparative fit index (CFI) of .95 or higher indicates a good fit (Marsh, Hau, & Wen, 2004). Analyses were performed for total adaptive ER and the different specific ER strategies separately. The latter analysis was performed because the specific adaptive ER strategies were highly correlated (see Table 1), thus creating a problem of multicollinearity.

Descriptive statistics

Means, standard deviations, and correlations between all variables are shown in Table 1. Mean CDI scores for the different age groups were M = 7.80 (SD = 6.10) for 7- to 12-year-olds with a mean of 7.90 (SD = 6.10) for girls and 7.70 (SD = 6.11) for boys, and M = 7.66 (SD = 5.79) for 13- to 17 year-olds with a mean of 8.13 (SD = 6.23) for girls and 7.29 (SD = 5.40) for boys, which compares favourably with values reported in previous studies using a community-based sample (Roelofs *et al.*, 2009; Vasey *et al.*, 2013). Mean CDI scores for the different groups can all be interpreted as average scores in comparison with peers (Kovacs, 1992; Timbremont & Braet, 2002). Furthermore, mean FEEL-KJ adaptive ER scores for the different age groups were M = 144.37 (SD = 28.95) for 7- to 12-year-olds with a mean of 142.56 (SD = 27.79) for girls and 146.00 (SD = 29.89) for boys, and

Table 1. Variable correlations and descriptives

Variables	1	2	3	4	5	6	7	8	9	10	*M* (*SD*)
1. PE											55.46 (8.14)
2. NE	−.14										33.49 (9.20)
3. EC	.32	−.33									48.28 (7.88)
4. CDI	−.44	.47	−.54								7.76 (6.00)
5. Adaptive ER	.41	−.19	.31	−.37							145.31 (28.86)
6. Reappraisal	.25	−.09	.14	−.22	.72						18.06 (5.69)
7. Problem-solving	.37	−.14	.33	−.34	.89	.54					42.21 (8.80)
8. Acceptance	.31	−.15	.23	−.29	.80	.49	.70				20.03 (4.80)
9. Positive refocusing	.36	−.18	.29	−.35	.81	.42	.65	.55			22.36 (5.45)
10. Distraction	.33	−.19	.20	−.28	.82	.44	.64	.62	.74		21.82 (5.66)
11. Forgetting	.36	.16	.30	−.34	.83	.63	.69	.59	.63	.58	20.84 (4.81)

Notes. NE, negative emotionality; PE, positive emotionality; EC, effortful control; CDI, depressive symptoms; Adaptive ER, total adaptive ER strategies.
N = 1,646, all correlations are significant at the $p < .001$ level.

$M = 147.30$ ($SD = 28.14$) for 13- to 17-year-olds with a mean of 146.41 ($SD = 27.90$) for girls and 148.00 ($SD = 28.35$) for boys. All mean scores corresponded with the average scores of peers (Braet *et al.*, 2013).

Preliminary analyses

Preliminary analysis using ANCOVA revealed that gender: $\beta = .09$, $t(1{,}638) = 4.23$, $p < .001$; age: $\beta = -.12$, $t(1{,}638) = -5.98$, $p < .001$; and EC: $\beta = -.58$, $t(1{,}638) = -28.66$, $p < .001$ were significantly related to depressive symptoms and were therefore controlled for in the structural models.

Measurement model

The estimated measurement models for reactive temperament traits, depressive symptoms, overall adaptive ER, and specific adaptive ER strategies yielded a good fit, χ^2s(48) = 239.94–302.87, $ps < .001$, all CFIs = .97, RMSEAs = .05 to .06, and SRMRs = .03 to .04.

Structural equation modelling

Reactive temperament traits and depressive symptoms

First, the *SEM* including the main effects of the reactive temperament traits on depressive symptoms showed a moderate fit, $\chi^2(44) = 428.88$, $p < .001$, CFI = .92, RMSEA = .07, SRMR = .04. As hypothesized, results supported a significant positive effect for NE ($\beta = .36$, $p < .001$) and a significant negative effect for PE ($\beta = -.37$, $p \leq .001$) on depressive symptoms.

Reactive temperament traits, adaptive ER, and depressive symptoms

Do total adaptive ER abilities mediate the relationship between PE and depressive symptoms?. The structural model with total adaptive ER as a mediator revealed a good fit, $\chi^2(76) = 653.03$, $p < .001$, CFI = .94, RMSEA = .07, SRMR = .04. As can be seen in Figure 1, the structural model proposes that both reactive temperament traits directly predicted total adaptive ER in the hypothesized direction and that total adaptive ER is associated with higher levels of depressive symptoms. However, to determine whether total adaptive ER significantly mediated the relationship between reactive temperament traits and depressive symptoms, we inspected the indirect effects using bootstrapped standard errors. Indirect effects were significant for the PE path ($\beta = -.04$, $p < .01$) but non-significant for the NE path ($\beta = .01$, $p = .06$). Results suggest that only PE yields a significant mediation effect through total adaptive ER on depressive symptoms. However, only partial mediation was obtained, given that the direct path from PE to depressive symptoms remained strongly significant.

Do all specific adaptive ER strategies contribute to further understanding the PE–depressive symptoms relationship?. In a second step, we explored the specific role of the various adaptive ER strategies in the reactive temperament–depressive symptoms relationship by testing a structural model for every specific adaptive ER strategy independently. All six models are shown in Figure 2.

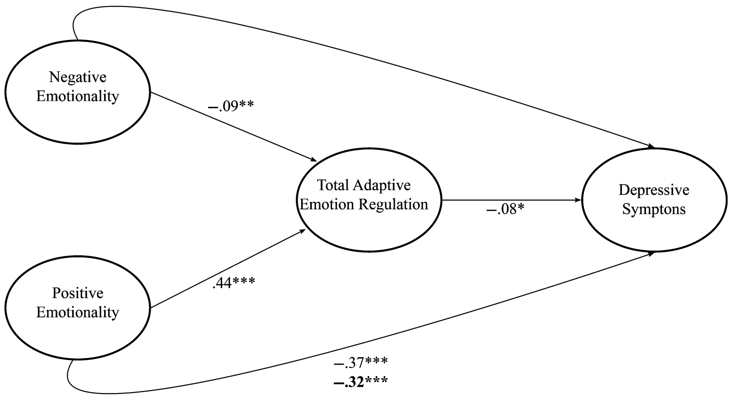

Figure 1. Structural model for the mediation model including total adaptive emotion regulation. Note: $*p < .05$, $**p < .01$, $***p < .001$.

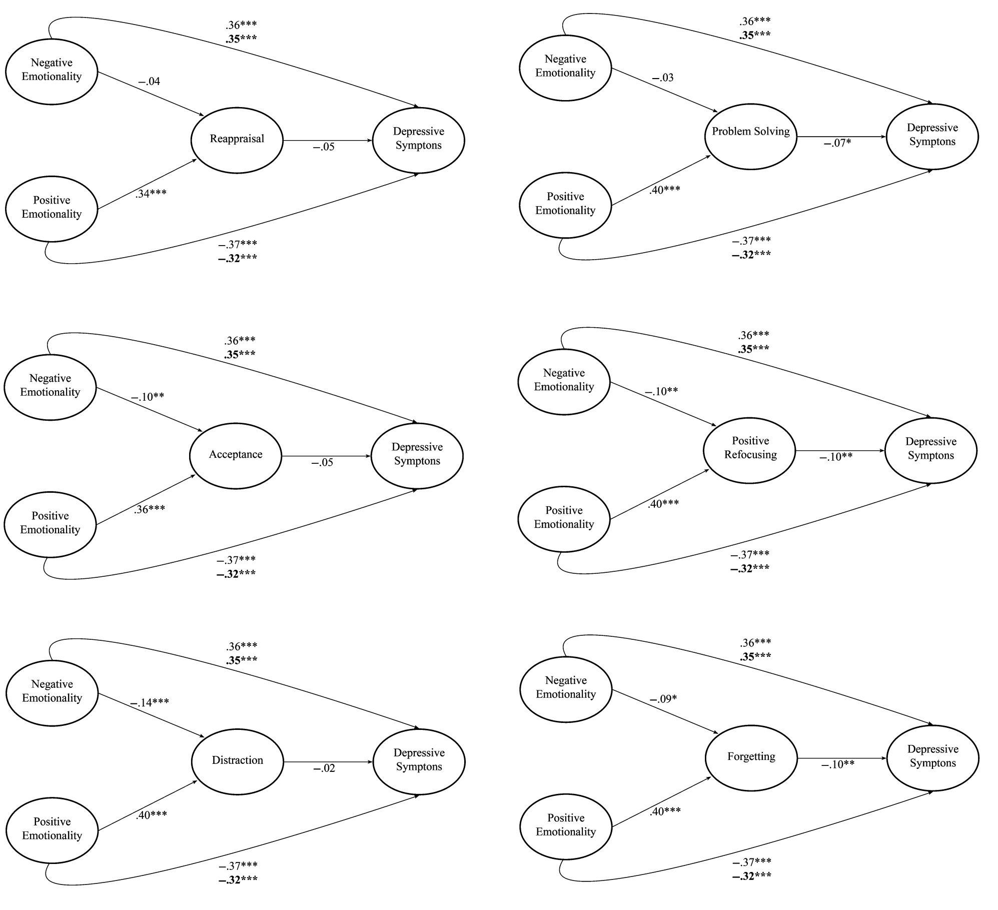

Figure 2. Structural models for the mediation models including specific adaptive emotion regulation strategies. Note: $*p < .05$, $**p < .01$, $***p < .001$.

First, the structural model pertaining to reappraisal showed an overall acceptable fit, $\chi^2(76) = 569.23$, $p < .001$, CFI = .94, RMSEA = .06, SRMR = .04 and suggested that PE, but not NE, predicted reappraisal. Indirect effects showed that reappraisal did not significantly mediate the relationship of PE and depressive symptomatology ($\beta = -.02$, $p = .06$), nor of NE ($\beta = .01$, $p = .37$) and such symptoms.

Second, the structural model including problem-solving showed an overall acceptable fit, $\chi^2(76) = 655.54$, $p < .001$, CFI = .94, RMSEA = .07, SRMR = .04 and also suggests that PE, but not NE, predicted problem-solving. Here, investigation of the indirect effects revealed that problem-solving significantly mediated the PE ($\beta = -.03$, $p < .05$)–depressive symptoms relationship, but not the NE ($\beta = .00$, $p = .45$)–depressive symptoms relationship. Notably, only partial mediation was obtained, given that the direct path from PE to depressive symptoms remained strongly significant.

Third, the structural model containing acceptance showed an overall acceptable fit, $\chi^2(76) = 620.11$, $p < .001$, CFI = .94, RMSEA = .07, SRMR = .04 and suggested that both PE and NE predicted acceptance. Indirect effects revealed that acceptance did not significantly mediate the PE ($\beta = -.02$, $p = .06$)–depressive symptoms relationship, nor the NE ($\beta = .00$, $p = .13$)–depressive symptoms relationship.

Fourth, the structural model pertaining positive refocusing showed an overall acceptable fit, $\chi^2(76) = 619.70$, $p < .001$, CFI = .94, RMSEA = .07, SRMR = .04 and suggested that both PE and NE predicted positive refocusing. Indirect effects revealed that positive refocusing significantly mediated the PE ($\beta = -.04$, $p < .01$)–depressive symptoms relationship and the NE ($\beta = .01$, $p < .05$)–depressive symptoms relationship. Moreover, only partial mediation was obtained, given that the direct paths from PE and NE to depressive symptoms remained strongly significant.

Next, the structural model pertaining distraction showed an overall acceptable fit, $\chi^2(76) = 593.63$, $p < .001$, CFI = .95, RMSEA = .06, SRMR = .04 and revealed that both PE and NE predicted distraction. Indirect effects revealed that distraction did neither significantly mediate the PE ($\beta = -.01$, $p = .56$)–depressive symptoms relationship nor the NE ($\beta = .00$, $p = .56$)–depressive symptoms relationship.

Lastly, the structural model pertaining forgetting showed an overall acceptable fit, $\chi^2(76) = 587.65$, $p < .001$, CFI = .94, RMSEA = .06, SRMR = .04 and showed that both PE and NE predicted forgetting. Indirect effects revealed that forgetting significantly mediated the PE ($\beta = -.04$, $p < .01$)–depressive symptoms relationship and the NE ($\beta = .01$, $p < .05$)–depressive symptoms relationship. However, only partial mediation was obtained, given that the direct paths from PE and NE to depressive symptoms remained strongly significant.

Moderating role of gender and age

Next, we tested the moderating role of gender and age for all relationships in the structural models through moderation analyses using latent variable interactions in Mplus (Muthén & Muthén, 2012). Results revealed that gender significantly moderated the relationship between NE and depressive symptoms ($b = .05$, $p < .01$). More specifically, high NE appeared to be associated with higher symptom levels in girls than in boys (see Figure 3). Furthermore, the relationship between NE and adaptive ER strategies was also significantly moderated by gender ($b = -2.40$, $p < .05$), showing that high NE was related to a lower use of adaptive ER strategies in girls, but not in boys (see Figure 4). Results revealed that the relationship between NE and total adaptive ER was also significantly moderated by age ($b = -.62$, $p = .05$). The simple slope became significant

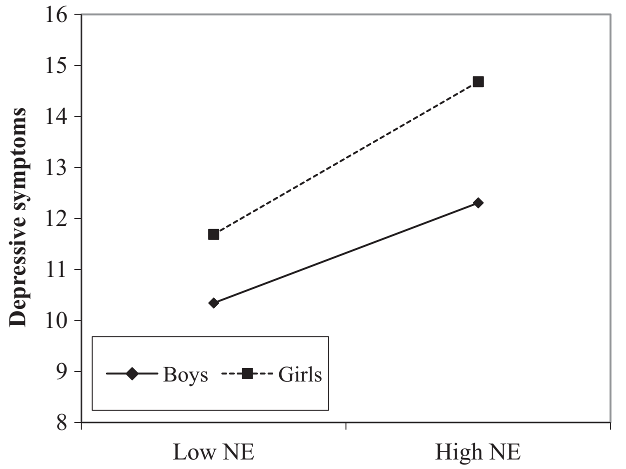

Figure 3. Interaction effect of negative emotionality (NE) and gender on depressive symptoms.

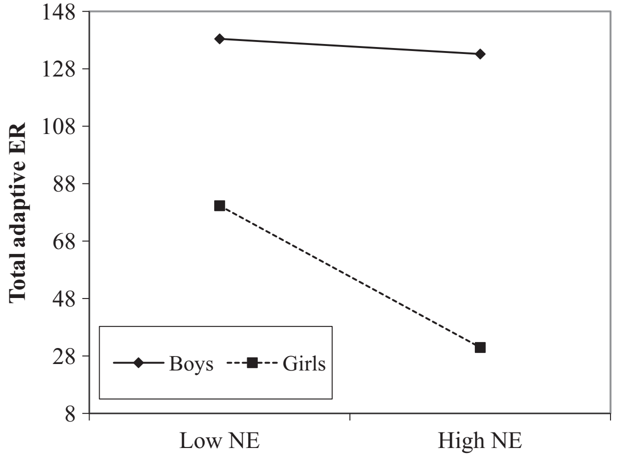

Figure 4. Interaction effect of negative emotionality (NE) and gender on total adaptive emotion regulation (ER).

starting at age 12 (see Figure 5), suggesting that from this age on higher NE was significantly associated with a lack of total adaptive ER. For the remaining mediation relationships pertaining total adaptive ER, no significant moderation was found. Similar moderation by gender and age was found in the models with the specific adaptive ER strategies (see Tables 2 and 3).

Discussion

The current study aimed to examine the unique relationships between temperament, adaptive ER strategies, and depression in youth using a cross-sectional design in a large community-based sample. Based on integrated cognitive-affective models on positive

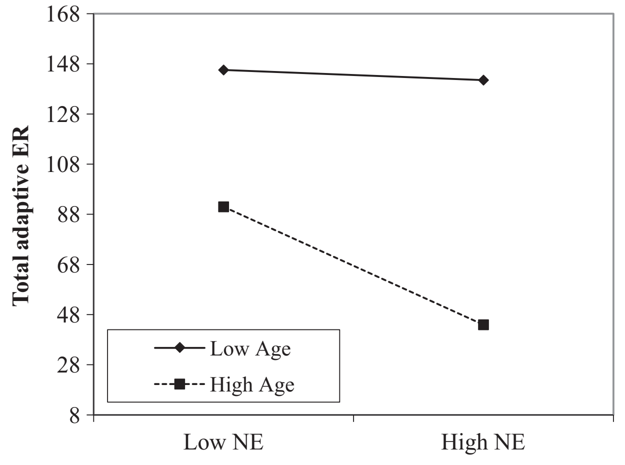

Figure 5. Interaction effect of negative emotionality (NE) and age on total adaptive emotion regulation (ER).

Table 2. Results of the moderated mediation analyses for gender for the different relations in the mediation models including specific adaptive emotion regulation strategies

	GEN × NE → CDI	GEN × PE → CDI	GEN × NE → ER	GEN × PE → ER	GEN × ER → CDI
Reappraisal	$b = .06$, $p < .01$	$b = -.02$, $p = .39$	$b = -.08$, $p = .42$	$b = -.04$, $p = .67$	$b = -.01$, $p = .45$
Problem-solving	$b = .06$, $p < .01$	$b = -.02$, $p = .32$	$b = -.15$, $p < .05$	$b = -.17$, $p < .05$	$b = .01$, $p = .82$
Acceptance	$b = .06$, $p < .01$	$b = -.02$, $p = .35$	$b = -.22$, $p < .01$	$b = -.04$, $p = .70$	$b = -.01$, $p = .33$
Positive refocusing	$b = .05$, $p < .01$	$b = -.02$, $p = .29$	$b = -.30$, $p < .001$	$b = -.11$, $p = .24$	$b = -.02$, $p = .15$
Distraction	$b = .06$, $p < .01$	$b = -.02$, $p = .41$	$b = -.23$, $p < .05$	$b = -.01$, $p = .93$	$b = -.01$, $p = .90$
Forgetting	$b = .06$, $p < .01$	$b = -.02$, $p = .32$	$b = -.12$, $p = .15$	$b = -.10$, $p = .21$	$b = -.01$, $p = .48$

Note. NE, negative emotionality; PE, positive emotionality; CDI, depressive symptoms; ER, specific emotion regulation strategy; GEN, gender.

affect and depression (Fredrickson, 2001; Hyde *et al.*, 2008), it was hypothesized that the relationship between PE and youth depressive symptoms would be mediated by adaptive ER strategies.

In support of our hypotheses, results showed that youth low in PE and youth high in NE reported greater depressive symptoms. However, while both reactive temperament traits appeared to be related to adaptive ER, which in turn was negatively related to depressive symptoms, only the PE–depressive symptoms relationship was significantly mediated by total adaptive ER abilities. While most research on temperament and depression has focused on identifying high NE as a temperamental vulnerability to depressive symptoms through the use of *maladaptive* ER strategies, such as rumination (Hankin *et al.*, 2009;

Mezulis *et al.*, 2010), the present findings indicate that low PE may similarly predict depressive symptoms through a lack of *adaptive* ER strategies (Harding *et al.*, 2014). This further supports the broaden-and-build theory (Fredrickson, 2001) suggesting that PE relates to the development of ER by predicting one's total adaptive ER abilities and thereby decreasing the risk of developing depressive symptoms (Tugade & Fredrickson, 2007). Specifically, PE may promote a tendency to broaden youth cognitive behavioural repertoires through the beneficial effects of one's reactivity to positive emotions, whereas NE is assumed to influence the development of overall maladaptive ER tendencies by narrowing one's thoughts and actions. Furthermore, a wider range of available adaptive ER strategies may offer youth greater flexibility to shift between different strategies, if necessary, and fine-tune their strategies to a specific situation (Bonanno, Papa, Lalande, Westphal, & Coifman, 2004). In adopting such a view, our results suggest that PE significantly impacts this flexibility.

Results also indicated that problem-solving, positive refocusing, and forgetting significantly mediated the relationship, whereas reappraisal, acceptance, and distraction did not. One explanation for these null findings is that the use of these strategies may be less related to reactive temperament traits. Our findings showed that the ER strategies most often implemented in therapeutic programmes targeting depression, that is *reappraisal* and *acceptance* (Stark & Kendall, 1996; Van der Gucht *et al.*, 2016), did not distinctly mediate the temperament–depressive symptoms relationship. Perhaps reappraisal and acceptance may need to be trained more explicitly as they are more complex and thus not automatically adopted in all youth (Hofmann, Schmeichel, & Baddeley, 2012; Schäfer *et al.*, 2016). PE was positively associated with reappraisal and acceptance, whereas acceptance was also negatively associated with NE. In following the idea that both ER strategies may need further training, one explanation could be that differences in reactive temperament traits may contribute to differences in the trainability of youth.

Distraction also did not contribute to the temperament–depression relationship. PE and NE were both associated with distraction, but problems with distraction by itself were

Table 3. Moderated mediation analyses of age and the different relations in the mediation models including specific adaptive emotion regulation strategies

	AGE × NE → CDI	AGE × PE → CDI	AGE × NE → ER	AGE × PE → ER	AGE × ER → CDI
Reappraisal	$b = .01$, $p = .50$	$b = .01$, $p = .62$	$b = -.03$, $p = .36$	$b = .01$, $p = .62$	$b = .00$, $p = .96$
Problem-solving	$b = .01$, $p = .56$	$b = .01$, $p = .68$	$b = -.03$, $p = .15$	$b = -.02$, $p = .40$	$b = -.01$, $p = .83$
Acceptance	$b = .01$, $p = .52$	$b = .01$, $p = .61$	$b = -.05$, $p < .05$	$b = .01$, $p = .69$	$b = -.01$, $p = .33$
Positive refocusing	$b = .01$, $p = .64$	$b = .01$, $p = .55$	$b = -.06$, $p < .01$	$b = .03$, $p = .26$	$b = -.01$, $p = .73$
Distraction	$b = .01$, $p = .55$	$b = .01$, $p = .69$	$b = -.07$, $p < .01$	$b = .03$, $p = .26$	$b = -.01$, $p = .60$
Forgetting	$b = .01$, $p = .59$	$b = .01$, $p = .73$	$b = -.04$, $p = .09$	$b = -.03$, $p = .21$	$b = -.01$, $p = .71$

Note. NE, negative emotionality; PE, positive emotionality; CDI, depressive symptoms; ER, specific emotion regulation strategy.

not significantly related to depressive symptoms. While distraction was successful in regulating negative affect short-term (Rood, Roelofs, Bögels, & Arntz, 2012), it may not assist youth in regulating negative affect long term because the stressors remain unresolved (Abela, Brozina, & Haigh, 2002; Abela, Vanderbilt, & Rochon, 2004). As a result, youth may need to shift to other ER strategies to regulate their negative affect. If youth choose to adopt a maladaptive ER strategy, instead of an adaptive one, this might heighten the risk for developing depressive symptoms. Possibly, differences in reactive temperament traits contribute to youth's preferred ER strategy choice. However, additional research on the serial and flexible use of different ER strategies is needed to further explore this hypothesis. We neither examined the activities involved in youth distraction nor whether these activities are adaptive (e.g., listening to music; Dingle, Hodges, & Kunde, 2016) or maladaptive (e.g., turning to drugs or alcohol; Nolen-Hoeksema & Harrell, 2002). As a result, distraction in our study may represent both adaptive and maladaptive ER.

Findings suggest that positive refocusing, forgetting, and problem-solving may represent more active agents in understanding why temperament heightens risk for developing youth depressive symptoms. While individuals frequently recruit pleasant memories to repair sad mood or negative affect, youth low in PE and high in NE may be less inclined to do so, thereby increasing the risk for developing depressive symptoms (Joormann, 2010). PE significantly contributed to the use of *positive refocusing*, which is consistent with research that has found that youth low in PE will be less likely to shift their attention to mood-incongruent (i.e., positive) information to repair their negative affect (Derryberry & Rothbart, 1997; Fredrickson, 2001). Positive refocusing also significantly mediated the NE–depressive symptoms relationship, which strengthens previous research that focusing on positive information can sometimes increase negative emotions (e.g., sadness), instead of successfully regulating them, among depressed individuals (Dunn, Dalgleish, Lawrence, Cusack, & Ogilvie, 2004). To successfully adopt this strategy, youth should also be able to downregulate negative emotions, which will be particularly difficult for youth high in NE.

Results further suggest that youth low in PE and youth high in NE will be less likely to intentionally try to forget negative events (Power, Dalgleish, Claudio, Tata, & Kentish, 2000), which puts these youth at heightened risk for developing depressive symptoms. *Forgetting* requires the youth to banish negative information from working memory (Joormann, 2010), which may be reflected by the significant relationship with NE. Difficulty expelling negative information from memory may eventually lead to the development of maladaptive ER strategies such as rumination in these youth (Joormann, 2010). However, PE also is significantly related to forgetting, which aligns with Law, Groome, Thorn, Potts, and Buchanan (2012) showing a significant positive correlation between retrieval-induced forgetting and extraversion among university students. Moreover, forgetting has been shown to increase positive emotions short-term by blocking or inhibiting retrieval of negative memories (Anderson, Bjork, & Bjork, 2000), which may be easier among youth high in PE (Fredrickson, 2001).

Third, *problem-solving* significantly mediated the relationship between PE and depressive symptoms but did not mediate the NE–symptoms relationship. In fact, NE did not directly predict problem-solving. This finding may be interpreted by the broaden-and-built theory, suggesting that positive emotions broaden habitual modes of thinking or acting, which may be of particular relevance to successfully adopt problem-solving as an ER strategy to regulate negative emotions (Fredrickson, 2001). Problem-solving involves specific actions directed at solving a problem such as brainstorming for solutions. Generating different solutions to a problem may require a broader mindset, which may be more available among youth high in PE. This finding is consistent with previous studies showing that positive emotions facilitate problem-solving abilities (Kaufmann, 2003).

Lastly, results revealed that the relationship between NE and depressive symptoms was stronger for girls than for boys. This finding may reflect the known gender difference in adolescent depression, showing that girls experience increases in depressive symptoms in early adolescence, while boys do not develop higher symptom levels until late adolescence (Salk, Petersen, Abramson, & Hyde, 2016). The relationship between NE and total adaptive ER also was moderated by gender. This finding may be explained within the ABC Model of the gender difference in adolescent depression (Hyde *et al.*, 2008) which asserts a stronger association between NE and ER for girls than for boys, but has thus far only been tested for maladaptive ER (Hyde *et al.*, 2008; Mezulis *et al.*, 2010). Our findings similarly support that NE may contribute to adaptive ER in girls but not boys. The NE–adaptive ER relationship also was significantly moderated by age, revealing that high NE was significantly related to lower adaptive ER abilities in youth starting from age 12. This finding may reflect the ongoing brain maturation of various cognitive abilities involved in ER during adolescence (McRae *et al.*, 2012; Steinberg, 2005). Consequently, more complex cognitive ER strategies (e.g., positive refocusing, acceptance) may be less accessible in the younger part of our sample as executive functions may not be fully developed yet (Ahmed, Bittencourt-Hewitt, & Sebastian, 2015; Steinberg, 2005).

Clinical implications

The clinical implications of the current study include greater emphasis on PE and adaptive ER abilities in the screening, prevention, and treatment of youth depression. We suggest that clinical practitioners fine-tune their therapeutic techniques to meet the temperamental profile of each depressed individual. While most treatment approaches focus on alleviating negative emotions, clinical practitioners should particularly aim at enhancing positive emotions in youth low in PE during treatment. Improving positive emotional functioning in the treatment of depressive symptoms and disorder may enhance resilience in addition to facilitating symptom reduction (Ehrenreich, Fairholme, Buzzella, Ellard, & Barlow, 2007).

Second, clinical practitioners may benefit from teaching youth other adaptive ER strategies aside from reappraisal. Such a view is consistent with previous research calling for incorporating other adaptive ER strategies, such as positive refocusing, into current CBT treatments (Hannesdottir & Ollendick, 2007). Findings suggest that youth low in PE will benefit from learning different adaptive ER skills, as multiple adaptive ER strategies might buffer them against negative effects related to their temperamental vulnerabilities.

Clinical practitioners also should consider EC when training youth to use an adaptive ER strategy (Van Beveren, McIntosh, *et al.*, 2016; Van Beveren, Mezulis, Wante, & Braet, 2016). Given that EC may play a central role in linking reactive temperament to adaptive ER (Fox & Calkins, 2003; Lonigan & Vasey, 2009; Verstraeten, Vasey, Raes, & Bijttebier, 2009), more extensive treatment may be needed for youth low in EC. Furthermore, additional training aimed at improving one's regulative abilities may be appropriate for some youth to ensure that they integrate the learned strategy into their ER repertoire. Findings underline the importance of pursuing a tailored approach in designing a treatment plan for youth depressive symptoms.

Limitations and future directions

Several limitations warrant discussion. First, the current study is limited by a cross-sectional design that cautions against causal inference. As a result, future research is needed to examine adaptive ER strategies longitudinally to investigate the developmental patterns of each strategy and their differential effectiveness across the developmental span.

Second, the current study exclusively depends on self-report measures. Such an approach can cause shared method variance, which may partially account for findings. Another potential concern of self-report studies involving temperament and depressive symptoms is that symptom levels may have influenced how youth answered items on the self-report measure assessing reactive temperament traits. For example, youth experiencing depressive symptoms may perceive themselves as experiencing more negative and less positive emotions in general as a result of their current emotional problems. For future research, we recommend implementing a multi-informant approach and combining multiple measurement techniques such as experimental tasks (e.g., Joormann & Gotlib, 2008), behavioural measures (Lonigan & Phillips, 2001), and physiological measures (for overview see Calkins & Swingler, 2012) to reliably assess temperament.

Third, as we used a community-based sample, levels of depressive symptoms were relatively low and did not reach the threshold for clinical depression for the majority of youth. Consequently, results should be cautiously generalized to clinically depressed youth. Future areas of research include investigating the proposed relationships in a clinical sample.

Fourth, youth may be less able to report what ER strategies they tend to use, as their capacity for introspection may not be fully developed (Eisenberg *et al.*, 2010). Although the FEEL-KJ has demonstrated good psychometric qualities with high internal consistency across all ages (Braet *et al.*, 2013), future research may benefit from using additional, multi-informant measures of adaptive ER strategies, as well as observational laboratory studies. On a related note, future studies also should include ER measures that consider what youth are doing when they are distracting themselves from negative emotions, to identify whether this ER strategy is indeed adopted in an adaptive way.

While the current study's primary focus was on the regulation of negative emotions (sadness, anger, and fear), further research is needed to examine the (dys)regulation of positive emotions and the detrimental effects of dysregulated positive emotions for developing depressive symptoms in youth (Carl *et al.*, 2013). A clearer understanding of how youth experience and regulate negative and positive emotions differently may provide a more integrative view on the risk of developing depressive symptoms during the adolescent period.

Lastly, our study used a trait-based approach to measure ER strategies. Future research also should investigate the use and the emotional consequences of the different ER strategies in daily life (e.g., Brans *et al.*, 2013). More specifically, we suggest that future studies measure emotion and ER strategies in daily life as close as possible to their occurrence while assessing the events eliciting these emotions.

To conclude, the current study supported low PE as a temperamental vulnerability for youth depressive symptoms, with PE being related to several adaptive ER abilities. Results add to the growing literature identifying cognitive and affective processes underlying the aetiology and maintenance of depressive symptoms in youth. Understanding these mechanisms is pivotal to the improvement of current prevention and treatment programmes.

References

Abela, J. R., Brozina, K., & Haigh, E. P. (2002). An examination of the response styles theory of depression in third-and seventh-grade children: A short-term longitudinal study. *Journal of Abnormal Child Psychology*, *30*, 515–527. https://doi.org/0091-0627/02/1000-0515/0

Abela, J. R. Z., Vanderbilt, E., & Rochon, A. (2004). A test of the integration of the response styles and social support theories of depression in third and seventh grade children. *Journal of Social and Clinical Psychology*, *23*, 653–674. https://doi.org/10.1521/jscp.23.5.653.50752

Ahmed, S. P., Bittencourt-Hewitt, A., & Sebastian, C. L. (2015). Neurocognitive bases of emotion regulation development in adolescence. *Developmental Cognitive Neuroscience*, *15*, 11–25. https://doi.org/10.1016/j.dcn.2015.07.006

Aldao, A., & Nolen-Hoeksema, S. (2012). When are adaptive strategies most predictive of psychopathology? *Journal of Abnormal Psychology*, *121*(1), 276–281. https://doi.org/10.1037/a0023598

Aldao, A., Nolen-Hoeksema, S., & Schweizer, S. (2010). Emotion-regulation strategies across psychopathology: A meta-analytic review. *Clinical Psychology Review*, *30*(2), 217–237. https://doi.org/10.1016/j.cpr.2009.11.004

American Psychiatric Association (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). Arlington, VA: Author.

Anderson, M. C., Bjork, E. L., & Bjork, R. A. (2000). Retrieval-induced forgetting: Evidence for a recall-specific mechanism. *Psychonomic Bulletin & Review*, *7*, 522–530.

Balázs, J., Miklósi, M., Keresztény, Á., Hoven, C. W., Carli, V., Wasserman, C., . . . Wasserman, D. (2013). Adolescent subthreshold depression and anxiety: Psychopathology, functional impairment and increased suicide risk. *Journal of Child Psychology and Psychiatry*, *54*, 670–677. https://doi.org/10.1111/jcpp.12016

Beck, A. T. (1970). Cognitive therapy – Nature and relation to behavior therapy. *Behavior Therapy*, *1*(2), 184–200. https://doi.org/10.1016/s0005-7894(70)80030-2

Beck, A. T., Rush, A. J., Shaw, B. F., & Emery, G. (1979). *Cognitive therapy of depression*. New York, NY: Guilford.

Birmaher, B., Ryan, N. D., Williamson, D. E., Brent, D. A., Kaufman, J., Dahl, R. E., . . . Nelson, B. (1996). Childhood and adolescent depression: A review of the past 10 years. Part I. *Journal of the American Academy of Child & Adolescent Psychiatry*, *35*, 1427–1439. https://doi.org/10.1097/00004583-199611000-00011

Bogels, S., Hoogstad, B., van Dun, L., de Schutter, S., & Restifo, K. (2008). Mindfulness training for adolescents with externalizing disorders and their parents. *Behavioural and Cognitive Psychotherapy*, *36*(2), 193–209. https://doi.org/10.1017/S1352465808004190

Bollen, K. A. (2014). *Structural equations with latent variables*. New York, NY: John Wiley & Sons.

Bonanno, G. A., Papa, A., Lalande, K., Westphal, M., & Coifman, K. (2004). The importance of being flexible the ability to both enhance and suppress emotional expression predicts long-term adjustment. *Psychological Science*, *15*, 482–487. https://doi.org/10.1111/j.0956-7976.2004.00705.x

Braet, C., Cracco, E., Theuwis, L., Grob, A., & Smolenski, C. (2013). *FEEL-KJ: Vragenlijst over emotieregulatie bij kinderen en jongeren [FEEL-KJ: Questionnaire for emotion regulation in children and adolescents]*. Amsterdam: Hogrefe.

Braet, C., Theuwis, L., Van Durme, K., Vandewalle, J., Vandevivere, E., Wante, L., . . . Goossens, L. (2014). Emotion regulation in children with emotional problems. *Cognitive Therapy and Research*, *38*, 493–504. https://doi.org/10.1007/s10608-014-9616-x

Brans, K., Koval, P., Verduyn, P., Lim, Y. L., & Kuppens, P. (2013). The regulation of negative and positive affect in daily life. *Emotion*, *13*, 926. https://doi.org/10.1037/a0032400

Burcusa, S. L., & Iacono, W. G. (2007). Risk for recurrence in depression. *Clinical Psychology Review*, *27*, 959–985. https://doi.org/10.1016/j.cpr.2007.02.005

Calkins, S. D., & Swingler, M. M. (2012). Psychobiological measures of temperament in childhood. In M. Zentner & R. L. Shiner (Eds.), *Handbook of temperament* (pp. 229–247). New York, NY: Guilford Press.

Carl, J. R., Soskin, D. P., Kerns, C., & Barlow, D. H. (2013). Positive emotion regulation in emotional disorders: A theoretical review. *Clinical Psychology Review*, *33*, 343–360. https://doi.org/10.1016/j.cpr.2013.01.003

Charles, S. T., Mather, M., & Carstensen, L. L. (2003). Aging and emotional memory: The forgettable nature of negative images for older adults. *Journal of Experimental Psychology: General*, *132*, 310–324. https://doi.org/10.1037/0096-3445.132.2.310

Clark, L. A., & Watson, D. (1991). Tripartite model of anxiety and depression – Psychometric evidence and taxonomic implications. *Journal of Abnormal Psychology*, *100*, 316–336. https://doi.org/10.1037/0021-843x.100.3.316

Copeland, W. E., Shanahan, L., Costello, E. J., & Angold, A. (2009). Childhood and adolescent psychiatric disorders as predictors of young adult disorders. *Archives of General Psychiatry*, *66*, 764–772. https://doi.org/10.1001/archgenpsychiatry.2009.85

Corr, P. J. (2002). JA Gray's reinforcement sensitivity theory and frustrative nonreward: A theoretical note on expectancies in reactions to rewarding stimuli. *Personality and Individual Differences*, *32*, 1247–1253. https://doi.org/10.1016/S0191-8869(01)00115-5

Craighead, W. E., Smucker, M. R., Craighead, L. W., & Ilardi, S. S. (1998). Factor analysis of the Children's Depression Inventory in a community sample. *Psychological Assessment*, *10*(2), 156. https://doi.org/10.1037/1040-3590.10.2.156

Derryberry, D., & Rothbart, M. K. (1997). Reactive and effortful processes in the organization of temperament. *Development and Psychopathology*, *9*, 633–652. https://doi.org/10.1017/S0954579497001375

Dingle, G. A., Hodges, J., & Kunde, A. (2016). Tuned in emotion regulation program using music listening: effectiveness for adolescents in educational settings. *Frontiers in Psychology*, *7*, 859–868. https://doi.org/10.3389/fpsyg.2016.00859

Dunn, B. D., Dalgleish, T., Lawrence, A. D., Cusack, R., & Ogilvie, A. D. (2004). Categorical and dimensional reports of experienced affect to emotion-inducing pictures in depression. *Journal of Abnormal Psychology*, *113*, 654–660. https://doi.org/10.1037/0021-843X.113.4.654

D'Zurilla, T. J., & Nezu, A. M. (1999). *Problem-solving therapy: A social competence approach to clinical intervention*. New York, NY: Springer.

Ehrenreich, J. T., Fairholme, C. P., Buzzella, B. A., Ellard, K. K., & Barlow, D. H. (2007). The role of emotion in psychological therapy. *Clinical Psychology: Science and Practice*, *14*, 422–428. https://doi.org/10.1111/j.1468-2850.2007.00102.x

Eisenberg, N., Spinrad, T. L., Eggum, N. D., Silva, K. M., Reiser, M., Hofer, C., . . . Popp, T. (2010). Relations among maternal socialization, effortful control, and maladjustment in early childhood. *Development and Psychopathology*, *22*, 507–525. https://doi.org/10.1017/S0954579410000246

Ellis, L. K., & Rothbart, M. K. (2001). *Revision of the early adolescent temperament questionnaire*. Paper presented at the Poster presented at the 2001 biennial meeting of the society for research in child development, Minneapolis, Minnesota.

Eysenck, H. J. (1967). *The biological basis of personality*, Vol. 689. New York, NY: Transaction Publishers.

Fergusson, D. M., & Woodward, L. J. (2002). Mental health, educational, and social role outcomes of adolescents with depression. *Archives of General Psychiatry*, *59*(3), 225–231. https://doi.org/10.1001/archpsyc.59.3.225

Folkman, S., & Moskowitz, J. T. (2000). Positive affect and the other side of coping. *American Psychologist*, *55*, 647. https://doi.org/10.1037/0003-066X.55.6.647

Fox, N. A., & Calkins, S. D. (2003). The development of self-control of emotion: Intrinsic and extrinsic influences. *Motivation and Emotion*, *27*(1), 7–26. https://doi.org/10.1023/A:1023622324898

Fredrickson, B. L. (2001). The role of positive emotions in positive psychology: The broaden-and-build theory of positive emotions. *The American Psychologist*, *56*(3), 218–226. https://doi.org/10.1098/rstb.2004.1512

Garland, E. L., Fredrickson, B., Kring, A. M., Johnson, D. P., Meyer, P. S., & Penn, D. L. (2010). Upward spirals of positive emotions counter downward spirals of negativity: Insights from the broaden-and-build theory and affective neuroscience on the treatment of emotion dysfunctions and deficits in psychopathology. *Clinical Psychology Review*, *30*, 849–864. https://doi.org/10.1016/j.cpr.2010.03.002

Garnefski, N., & Kraaij, V. (2006). Relationships between cognitive emotion regulation strategies and depressive symptoms: A comparative study of five specific samples. *Personality and Individual Differences*, *40*, 1659–1669. https://doi.org/10.1016/j.paid.2005.12.009

Garnefski, N., Kraaij, V., & Spinhoven, P. (2001). Negative life events, cognitive emotion regulation and emotional problems. *Personality and Individual Differences*, *30*, 1311–1327. https://doi.org/10.1016/s0191-8869(00)00113-6

Gilbert, K. E. (2012). The neglected role of positive emotion in adolescent psychopathology. *Clinical Psychology Review*, *32*, 467–481. https://doi.org/10.1016/j.cpr.2012.05.005

Grob, A., & Smolenski, C. (2005). *Fragebogen zur Erhebung der Emotionsregulation bei Kindern und Jugendlichen (FEEL-KJ)*. Bern: Verlag Hans Huber.

Gross, J. J. (1998). Antecedent- and response-focused emotion regulation: Divergent consequences for experience, expression, and physiology. *Journal of Personality and Social Psychology*, *74*(1), 224–237. https://doi.org/10.1037//0022-3514.74.1.224

Gross, J. J. (1999). Emotion regulation: Past, present, future. *Cognition & Emotion*, *13*, 551–573. https://doi.org/10.1080/026999399379186

Gross, J. J., & Thompson, R. A. (2007). *Emotion regulation: Conceptual foundations*. New York, NY: Guilford Press

Hankin, B. L., Oppenheimer, C., Jenness, J., Barrocas, A., Shapero, B. G., & Goldband, J. (2009). Developmental origins of cognitive vulnerabilities to depression: Review of processes contributing to stability and change across time. *Journal of Clinical Psychology*, *65*, 1327–1338. https://doi.org/10.1002/jclp.20625

Hannesdottir, D. K., & Ollendick, T. H. (2007). The role of emotion regulation in the treatment of child anxiety disorders. *Clinical Child and Family Psychology Review*, *10*(3), 275–293. https://doi.org/10.1007/s10567-007-0024-6

Harding, K. A., Hudson, M. R., & Mezulis, A. (2014). Cognitive mechanisms linking low trait positive affect to depressive symptoms: A prospective diary study. *Cognition & Emotion*, *28*, 1502–1511. https://doi.org/10.1080/02699931.2014.889661

Hartman, C. (2000). Nederlandse vertaling van de Early Adolescent Temperament Questionnaire. *Intern rapport, Disciplinegroep Psychiatrie*. *[Dutch translation of the Early Adolescent Temperament Questionnaire. Internal report, Discipline group Psychiatry]* Nederland: Universiteit van Groningen.

Heffner, M., Eifert, G. H., Parker, B. T., Hernandez, D. H., & Sperry, J. A. (2003). Valued directions: Acceptance and commitment therapy in the treatment of alcohol dependence. *Cognitive and Behavioral Practice*, *10*, 378–383. https://doi.org/10.1016/s1077-7229(03)80055-x

Henry, J. D., Castellini, J., Moses, E., & Scott, J. G. (2016). Emotion regulation in adolescents with mental health problems. *Journal of Clinical and Experimental Neuropsychology*, *38*(2), 197–207. https://doi.org/10.1080/13803395.2015.1100276

Hilt, L. M., McLaughlin, K. A., & Nolen-Hoeksema, S. (2010). Examination of the response styles theory in a community sample of young adolescents. *Journal of Abnormal Child Psychology*, *38*, 545–556. https://doi.org/10.1007/s10802-009-9384-3

Hofmann, W., Schmeichel, B. J., & Baddeley, A. D. (2012). Executive functions and self-regulation. *Trends in Cognitive Sciences*, *16*(3), 174–180. https://doi.org/10.1016/j.tics.2012.01.006

Hudson, M. R., Harding, K. A., & Mezulis, A. (2015). Dampening and brooding jointly link temperament with depressive symptoms: A prospective study. *Personality and Individual Differences*, *83*, 249–254. https://doi.org/10.1016/j.paid.2015.04.025

Hyde, J. S., Mezulis, A. H., & Abramson, L. Y. (2008). The ABCs of depression: Integrating affective, biological, and cognitive models to explain the emergence of the gender difference in depression. *Psychological Review*, *115*(2), 291–313. https://doi.org/10.1037/0033-295X.115.2.291

Jimenez, S. S., Niles, B. L., & Park, C. L. (2010). A mindfulness model of affect regulation and depressive symptoms: Positive emotions, mood regulation expectancies, and self-acceptance as regulatory mechanisms. *Personality and Individual Differences*, *49*, 645–650. https://doi.org/10.1016/j.paid.2010.05.041

Joormann, J. (2010). Cognitive inhibition and emotion regulation in depression. *Current Directions in Psychological Science*, *19*(3), 161–166. https://doi.org/10.1177/0963721410370293

Joormann, J., & Gotlib, I. H. (2008). Updating the contents of working memory in depression: Interference from irrelevant negative material. *Journal of Abnormal Psychology*, *117*(1), 182. https://doi.org/10.1037/0021-843X.117.1.182

Just, N., & Alloy, L. B. (1997). The response styles theory of depression: Tests and an extension of the theory. *Journal of Abnormal Psychology*, *106*(2), 221–229. https://doi.org/10.1037/0021-843x.106.2.221

Kaufmann, G. (2003). The effect of mood on creativity in the innovative process. *International handbook on innovation*, *1*, 191–203.

Kennedy, Q., Mather, M., & Carstensen, L. L. (2004). The role of motivation in the age-related positivity effect in autobiographical memory. *Psychological Science*, *15*(3), 208–214. https://doi.org/10.1111/j.0956-7976.2004.01503011.x

Kovacs, M. (1992). *Children's depression inventory (CDI) manual*. New York, NY: Multi-Health Systems.

Laurent, J., Catanzaro, S. J., Joiner, Jr, T. E., Rudolph, K. D., Potter, K. I., Lambert, S., … Gathright, T. (1999). A measure of positive and negative affect for children: Scale development and preliminary validation. *Psychological Assessment*, *11*, 326. https://doi.org/10.1037/1040-3590.11.3.326

Law, R., Groome, D., Thorn, L., Potts, R., & Buchanan, T. (2012). The relationship between retrieval-induced forgetting, anxiety, and personality. *Anxiety Stress and Coping*, *25*, 711–718. https://doi.org/10.1080/10615806.2011.630070

Lewinsohn, P. (1975). Engagement in pleasant activities and depression level. *Journal of Abnormal Psychology*, *84*(6), 729–731. https://doi.org/10.1037/0021-843X.84.6.729

Little, R. J. (1988). A test of missing completely at random for multivariate data with missing values. *Journal of the American Statistical Association*, *83*, 1198–1202. https://doi.org/10.1080/01621459.1988.10478722

Lonigan, C. J., & Phillips, B. M. (2001). Temperamental influences on the development of anxiety disorders. In M. W. Vasey & M. R. Dadds (Eds.), *The developmental psychopathology of anxiety* (pp. 60–91). New York, NY: Oxford University Press.

Lonigan, C. J., Phillips, B. M., & Hooe, E. S. (2003). Relations of positive and negative affectivity to anxiety and depression in children: Evidence from a latent variable longitudinal study. *Journal of Consulting and Clinical Psychology*, *71*, 465. https://doi.org/10.1037/0022-006X.71.3.465

Lonigan, C. J., & Vasey, M. W. (2009). Negative affectivity, effortful control, and attention to threat-relevant stimuli. *Journal of Abnormal Child Psychology*, *37*, 387–399. https://doi.org/10.1007/s10802-008-9284-y

Marroquín, B. (2011). Interpersonal emotion regulation as a mechanism of social support in depression. *Clinical Psychology Review*, *31*, 1276–1290. https://doi.org/10.1016/j.cpr.2011.09.005

Marsh, H. W., Hau, K.-T., & Wen, Z. (2004). In search of golden rules: Comment on hypothesis-testing approaches to setting cutoff values for fit indexes and dangers in overgeneralizing Hu and Bentler's (1999) findings. *Structural Equation Modeling*, *11*, 320–341. https://doi.org/10.1207/s15328007sem1103_2

McRae, K., Gross, J. J., Weber, J., Robertson, E. R., Sokol-Hessner, P., Ray, R. D., … & Ochsner, K. N. (2012). The development of emotion regulation: an fMRI study of cognitive reappraisal in children, adolescents and young adults *Social Cognitive and Affective Neuroscience*, 7(1), 11–22. https://doi.org/10.1093/scan/nsr093

Mehrabi, A., Mohammadkhani, P., Dolatshahi, B., Pourshahbaz, A., & Mohammadi, A. (2014). Emotion regulation in depression: An integrative review. *Practice in Clinical Psychology*, *2*(3), 181–194.

Mezulis, A. H., Priess, H. A., & Hyde, J. S. (2010). Rumination mediates the relationship between infant temperament and adolescent depressive symptoms. *Depression research and treatment, 2011*. https://doi.org/10.1155/2011/487873

Muris, P., & Meesters, C. (2009). Reactive and regulative temperament in youths: Psychometric evaluation of the early adolescent temperament questionnaire-revised. *Journal of Psychopathology and Behavioral Assessment*, *31*(1), 7–19. https://doi.org/10.1007/s10862-008-9089-x

Muthén, L., & Muthén, B. (2012). *Mplus statistical modeling software: Release 7.0*. Los Angeles, CA: Author.

Nock, M. K., Green, J. G., Hwang, I., McLaughlin, K. A., Sampson, N. A., Zaslavsky, A. M., & Kessler, R. C. (2013). Prevalence, correlates, and treatment of lifetime suicidal behavior among adolescents results from the National Comorbidity Survey Replication Adolescent Supplement. *JAMA Psychiatry*, *70*, 300–310. https://doi.org/10.1001/2013.jamapsychiatry.55

Nolen-Hoeksema, S. (1991). Responses to depression and their effects on the duration of depressive episodes. *Journal of Abnormal Psychology*, *100*, 569–582. https://doi.org/10.1037/0021-843x.100.4.569

Nolen-Hoeksema, S., & Harrell, Z. A. (2002). Rumination, depression, and alcohol use: Tests of gender differences. *Journal of Cognitive Psychotherapy*, *16*, 391–403. https://doi.org/10.1891/jcop.16.4.391.52526

Nolen-Hoeksema, S., & Morrow, J. (1993). Effects of rumination and distraction on naturally-occurring depressed mood. *Cognition & Emotion*, *7*, 561–570. https://doi.org/10.1080/02699939308409206

Nolen-Hoeksema, S., Wisco, B. E., & Lyubomirsky, S. (2008). Rethinking rumination. *Perspectives on Psychological Science*, *3*, 400–424. https://doi.org/10.1111/j.1745-6924.2008.00088.x

Power, M. J., Dalgleish, T., Claudio, V., Tata, P., & Kentish, J. (2000). The directed forgetting task: Application to emotionally valent material. *Journal of Affective Disorders*, *57*(1–3), 147–157. https://doi.org/10.1016/s0165-0327(99)00084-1

Putnam, S. P., Ellis, L. K., & Rothbart, M. K. (2001). The structure of temperament from infancy through adolescence. In A. Eliasz & Angleitner (Eds.), *Advances in Research on Temperament* (pp. 165–182). Lengerich: Pabst Scientist Publisher.

Roelofs, J., Rood, L., Meesters, C., te Dorsthorst, V., Bogels, S., Alloy, L. B., & Nolen-Hoeksema, S. (2009). The influence of rumination and distraction on depressed and anxious mood: A prospective examination of the response styles theory in children and adolescents. *European Child & Adolescent Psychiatry*, *18*, 635–642. https://doi.org/10.1007/s00787-009-0026-7

Rood, L., Roelofs, J., Bögels, S. M., & Arntz, A. (2012). The effects of experimentally induced rumination, positive reappraisal, acceptance, and distancing when thinking about a stressful event on affect states in adolescents. *Journal of Abnormal Child Psychology*, *40*(1), 73–84. https://doi.org/10.1007/s10802-011-9544-0

Rothbart, M. K., Ahadi, S. A., & Evans, D. E. (2000). Temperament and personality: Origins and outcomes. *Journal of Personality and Social Psychology*, *78*(1), 122–135. https://doi.org/10.1037//0022-3514.78.1.122

Salk, R. H., Petersen, J. L., Abramson, L. Y., & Hyde, J. S. (2016). The contemporary face of gender differences and similarities in depression throughout adolescence: Development and chronicity. *Journal of Affective Disorders*, *205*, 28–35. https://doi.org/10.1016/j.jad.2016.03.071

Samson, A. C., & Gross, J. J. (2012). Humour as emotion regulation: The differential consequences of negative versus positive humour. *Cognition & Emotion*, *26*, 375–384. https://doi.org/10.1080/02699931.2011.585069

Schafer, J. L. (1997). *Analysis of incomplete multivariate data*. London: Chapman & Hall.

Schäfer, J. Ö., Naumann, E., Holmes, E. A., Tuschen-Caffier, B., & Samson, A. C. (2016). Emotion regulation strategies in depressive and anxiety symptoms in youth: A meta-analytic review. *Journal of Youth and Adolescence*, *46*(2), 1–16.

Schmitt, K., Gold, A., & Rauch, W. (2012). Deficient adaptive regulation of emotion in children with ADHD. *Zeitschrift fur Kinder-und Jugendpsychiatrie und Psychotherapie*, *40*(2), 95–102; quiz 102-103. https://doi.org/10.1024/1422-4917/a000156

Sheppes, G., Scheibe, S., Suri, G., & Gross, J. J. (2011). Emotion-regulation choice. *Psychological Science*, *22*, 1391–1396. https://doi.org/10.1177/0956797611418350

Stark, K. D., & Kendall, P. C. (1996). *Treating depressed children: Therapist manual for taking action*. Ardmore, PA: Workbook Publishing.

Steinberg, L. (2005). Cognitive and affective development in adolescence. *Trends in Cognitive Sciences*, *9*(2), 69–74. https://doi.org/10.1016/j.tics.2004.12.00

Teasdale, J. D., Segal, Z. V., Williams, J. M. G., Ridgeway, V. A., Soulsby, J. M., & Lau, M. A. (2000). Prevention of relapse/recurrence in major depression by mindfulness-based cognitive therapy. *Journal of Consulting and Clinical Psychology*, *68*, 615–623. https://doi.org/10.1037//0022-006x.68.4.615

Thayer, J. F., Rossy, L. A., Ruiz-Padial, E., & Johnsen, B. H. (2003). Gender differences in the relationship between emotional regulation and depressive symptoms. *Cognitive Therapy and Research*, *27*, 349–364. https://doi.org/10.1023/A:1023922618287

Timbremont, B., & Braet, C. (2002). *Children's Depression Inventory, Nederlandstalige versie (Children's Depression Inventory, Dutch version)*. Lisse: Swets Test Publishers.

Tugade, M. M., & Fredrickson, B. L. (2007). Regulation of positive emotions: Emotion regulation strategies that promote resilience. *Journal of Happiness Studies*, *8*, 311–333. https://doi.org/10.1007/s10902-006-9015-4

Van Beveren, M.-L., McIntosh, K., Vandevivere, E., Wante, L., Vandeweghe, L., Van Durme, K., . . . Braet, C. (2016). Associations between temperament, emotion regulation, and depression in youth: The role of positive temperament. *Journal of Child and Family Studies*, *25*, 1954–1968. https://doi.org/10.1007/s10826-016-0368-y

Van Beveren, M. L., Mezulis, A., Wante, L., & Braet, C. (2016). Joint contributions of negative emotionality, positive emotionality, and effortful control on depressive symptoms in youth. *Journal of Clinical Child & Adolescent Psychology*, *25*(6), 1954–1968. https://doi.org/10.1080/15374416.2016.1233499

Van der Gucht, K., Kuppens, P., Maex, E., & Raes, F. (2016). Mindfulness-based interventions as school-based mental health promoting programs. In *Mindfulness-based cognitive therapy* (pp. 229–235). Cham, Switzerland: Springer International Publishing.

Van Dillen, L. F., & Papies, E. K. (2015). From distraction to mindfulness: Psychological and neural mechanisms of attention strategies in self-regulation. In R.B. Lopez, K.D. Vohs, D.D, Wagner & T. F, Heatherton (Ed.), *Handbook of biobehavioral approaches to self-regulation* (pp. 141–154). New York, NY: Springer.

Vasey, M. W., Harbaugh, C. N., Fisher, L. B., Heath, J. H., Hayes, A. F., & Bijttebier, P. (2014). Temperament synergies in risk for and protection against depressive symptoms: A prospective replication of a three-way interaction. *Journal of Research in Personality*, *53*, 134–147. https://doi.org/10.1016/j.jrp.2014.09.005

Vasey, M. W., Harbaugh, C. N., Lonigan, C. J., Phillips, B. M., Hankin, B. L., Willem, L., & Bijttebier, P. (2013). Dimensions of temperament and depressive symptoms: Replicating a three-way interaction. *Journal of Research in Personality*, *47*, 908–921. https://doi.org/10.1016/j.jrp.2013.09.001

Verstraeten, K., Vasey, M. W., Raes, F., & Bijttebier, P. (2009). Temperament and risk for depressive symptoms in adolescence: Mediation by rumination and moderation by effortful control. *Journal of Abnormal Child Psychology*, *37* 349–361. https://doi.org/10.1007/s10802-008-9293-x

Verstraeten, K., Vasey, M. W., Raes, F., & Bijttebier, P. (2012). The mediational role of responses to positive affect in the association between temperament and (hypo) manic symptoms in children. *Cognitive Therapy and Research*, *36*, 768–775. https://doi.org/10.1007/s10608-011-9402-y

Williams, J. M. G., Teasdale, J. D., Segal, Z. V., & Soulsby, J. (2000). Mindfulness-based cognitive therapy reduces overgeneral autobiographical memory in formerly depressed patients. *Journal of Abnormal Psychology*, *109*(1), 150–155. https://doi.org/10.1037//0021-843x.109.1.150

Yap, M. B. H., Allen, N. B., O'Shea, M., Di Parsia, P., Simmons, J. G., & Sheeber, L. (2011). Early adolescents' temperament, emotion regulation during mother-child interactions, and depressive symptomatology. *Development and Psychopathology*, *23*(1), 267–282. https://doi.org/10.1017/S0954579410000787

Yap, M. B. H., Allen, N. B., & Sheeber, L. (2007). Using an emotion regulation framework to understand the role of temperament and family processes in risk for adolescent depressive disorders. *Clinical Child and Family Psychology Review*, *10*(2), 180–196. https://doi.org/10.1007/s10567-006-0014-0

Received 23 December 2016; revised version received 21 June 2017